Matrix 2 tutorial 6

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1 Double Integrals

A double integral represents the volume under a surface $f(x, y)$ over a region D:

$$\iint_D f(x, y)\, dA$$

It can be computed as an iterated integral:

$$\iint_D f(x, y)\, dA = \int \int f(x, y)\, dx\, dy$$

Problems

1. $$\int_0^2 \int_0^3 (x + 2y)\, dy\, dx$$

2. $$\int_0^1 \int_0^x (x^2 + y^2)\, dy\, dx$$

3. $$\int_0^2 \int_x^{2x} xy\, dy\, dx$$

2 Domains and Changing Order of Integration

The limits of integration depend on the geometry of the domain. Sketching the region is essential.

Problems

1. $$\int_0^1 \int_x^1 1\, dy\, dx$$

2. Reverse the order:
$$\int_0^1 \int_y^1 f(x,y)\,dx\,dy$$

3. Reverse the order and evaluate:
$$\int_0^4 \int_{\sqrt{x}}^2 1\,dy\,dx$$

3 Polar Coordinates

For circular regions, use:

$$x = r\cos\theta, \quad y = r\sin\theta$$

Area element transforms as:

$$dA = r\,dr\,d\theta$$

Problems

1.
$$\iint_D (x^2+y^2)\,dA, \quad x^2+y^2 \le 4$$

2.
$$\iint_D x\,dA, \quad x^2+y^2 \le 1,\ y \ge 0$$

3.
$$\iint_{R^2} e^{-(x^2+y^2)}\,dA$$

4 Cylindrical Coordinates

Transformation:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$$

Volume element:

$$dV = r\,dr\,d\theta\,dz$$

Problems

1.
$$\iiint_V r\,dV, \quad r \le 2,\ 0 \le z \le 3$$

2.
$$\iiint_V z\,dV, \quad r \le 1,\ 0 \le z \le 4 - r^2$$

5 Spherical Coordinates

Transformation:

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho \cos\phi$$

Volume element:

$$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

Problems

(a)

$$\iiint_V 1 \, dV, \quad \text{sphere of radius 3}$$

(b)

$$\iiint_V z \, dV, \quad x^2 + y^2 + z^2 \leq 4, \ z \geq 0$$

(c)

$$\iiint_V (x^2 + y^2 + z^2) \, dV, \quad \rho \leq R$$

6 Change of Variables and the Jacobian

6.1 Theorem

Let (x, y) be given by a transformation:

$$x = x(u, v), \quad y = y(u, v)$$

Then:

$$\iint_D f(x, y) \, dx \, dy = \iint_{D'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du \, dv$$

where the Jacobian is:

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

6.2 Conclusion

$$dx \, dy = |J| \, du \, dv$$

7 Exercises: Curvilinear and Cylindrical Coordinates

(a)
$$\iint_D (x^2 + y^2)\, dA, \quad D: \ x^2 + y^2 \leq 9$$

(b)
$$\iiint_V 1\, dV, \quad V: \ r \leq 2, \ 0 \leq z \leq 5$$

(c)
$$\iiint_V z\, dV, \quad V: \ 0 \leq r \leq 1, \ 0 \leq \theta \leq 2\pi, \ 0 \leq z \leq 2$$

(d)
$$\iiint_V r^2\, dV, \quad V: \ r \leq 3, \ 0 \leq z \leq r$$

(e)
$$\iint_D (x^2 + y^2)\, dA, \quad D: \ y = 0, \ x = 2, \ x = y$$

(f)
$$\iiint_V z\, dV, \quad V: \ 0 \leq z \leq 4 - r^2$$

(g)
$$\iiint_V (x^2 + y^2 + z^2)\, dV, \quad V: \ x^2 + y^2 \leq 4, \ 0 \leq z \leq x^2 + y^2$$

(h)
$$\iiint_V e^{-(x^2+y^2)}\, dV, \quad V: \ x^2 + y^2 < \infty, \ 0 \leq z \leq 1$$

(i) Find the volume:
$$z = r, \quad z = 4, \quad 0 \leq \theta \leq 2\pi$$